

Lab Experiment 13 (Open-ended Lab)

Objective: To *build* a 5-stage RISC-V pipelined processor capable of executing provided assembly instructions.

Background

The speed of a system is characterized by the latency and throughput of information moving through it. We define a *token* to be a group of inputs that are processed to produce a group of outputs. The term conjures up the notion of placing subway tokens on a circuit diagram and moving them around to visualize data moving through the circuit. The *latency* of a system is the time required for one token to pass through the system from start to end. The *throughput* is the number of tokens that can be produced per unit time. As you might imagine, the throughput can be improved by processing several tokens at the same time. This is called *parallelism*, which comes in two forms: spatial and temporal. With *spatial parallelism*, multiple copies of the hardware are provided so that multiple tasks can be done at the same time. With *temporal parallelism*, a task is broken into stages, like an assembly line. Multiple tasks can be spread across the stages. Although each task must pass through all stages, a different task will be in each stage at any given time, so multiple tasks can overlap. Temporal parallelism is commonly called *pipelining*. Spatial parallelism is sometimes just called *parallelism*.

We design a pipelined processor by subdividing the single-cycle processor into five pipeline stages. Thus, five instructions can execute simultaneously, one in each stage. Because each stage has only one-fifth of the entire logic, the clock frequency is approximately five times faster. So, ideally, the latency of each instruction is unchanged, but the throughput is five times better. Microprocessors execute millions or billions of instructions per second, so throughput is more important than latency. Figure 1 shows 5 stage pipelined RISC-V processor with control signals. Write Verilog implementation of provided RISC-V pipelined processor.

Submission Deliverables:

1. The Verilog files for modules as well as test benches module wise.
2. A report containing explanations of how you implemented your processor, test cases and results, any difficulties you had and how you overcame them, and any deficiencies in your project if there are any. Also, add all the codes in appendices.

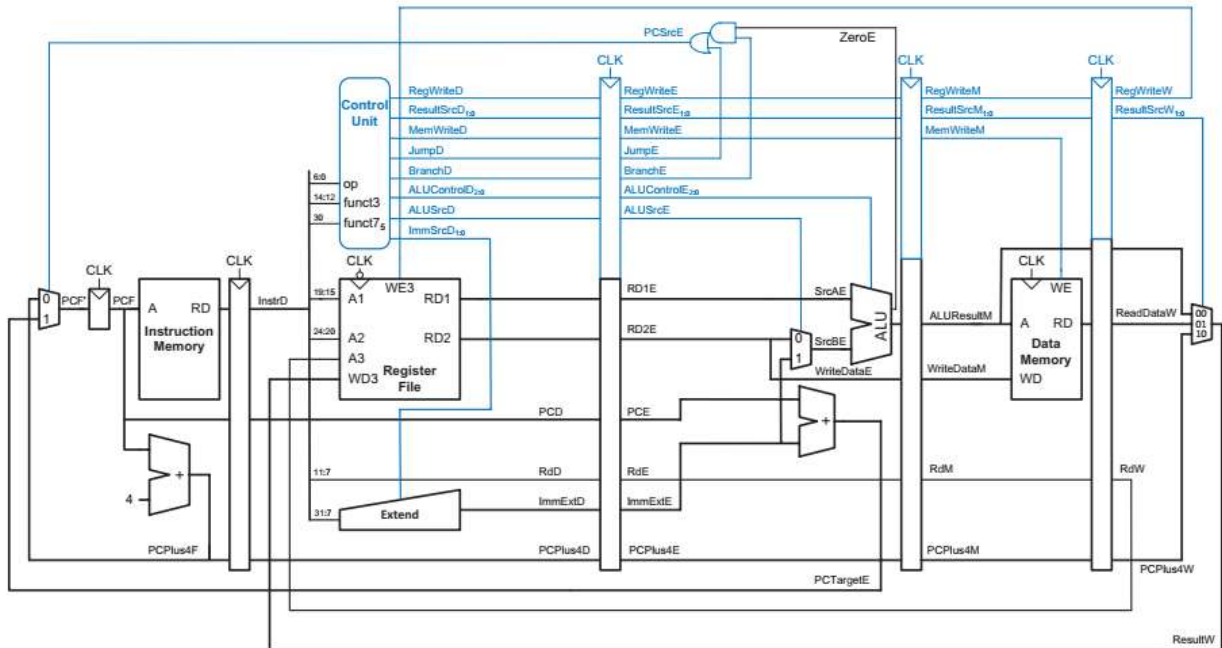


Figure 1-Pipelined processor with Control

Tasks:

1. Initialize registers with following values: x2 – 5, x3 – 12, x4 – 3
2. Feed following assembly instruction's machine code in Instruction memory

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Main:  addi x7, x3, -9
       sw x4, x3(0)
       or x4, x7, x2
       and x5, x3, x4
       beq x5, x7, end
       lw x3, x3(0)
       slt x4, x4, x3
       beq x4, x0, around
around: beq x2, x2, around
end:   add x3, x4, x2
  
```

3. Implement pipeline RISC-V processor (with hazard unit, if required).
4. Discuss hazards involved in the execution of above mentioned instructions, if any.

Reference

Harris, D., & Harris, S. (2010). Digital design and computer architecture. Morgan Kaufmann.

Associated GA, WK and CEA attributes

GA/ WA	Knowledge Profile (WK)	CEA Attributes
☐ WA1 ☐ WA2 ☐ WA3 ☐ WA4 ☒ WA5 ☐ WA6 ☐ WA7 ☐ WA8 ☐ WA9 ☐ WA10 ☐ WA11 ☐ WA12	☐ WK1: Natural Sciences ☐ WK2: Mathematics and Computing ☐ WK3: Engineering Fundamentals ☐ WK4: Engineering Specializations ☐ WK5: Engineering Design ☒ WK6: Engineering Practices ☐ WK7: Engineering in Society ☐ WK8: Research Literature	☐ Range of Resources ☒ Level of interaction ☐ Innovation ☐ Consequence to society and the environment ☐ Familiarity



F/OBEM 01/18/00

NED University of Engineering & Technology
Department of **Electronic** Engineering
Course Code and Title: EL - 421 Embedded Electronics

Laboratory Session No. _____

Date: _____

Psychomotor Domain Assessment Rubric-Level P3					
Skill Sets	Extent of Achievement				
	0	1	2	3	4
<u>Pipelined processor implementation</u>	It is not implemented.	Incorrect logic/pipeline register modules.	Incomplete implementation /few conditions are not included.	It is implemented with few errors.	It is correctly implemented.
<u>Hazard Unit</u>	It is not implemented.	Incorrect logic for the module.	Incomplete implementation /few conditions are not included.	It is implemented with certain errors.	It is correctly implemented.
<u>Functionality</u>	The processor is not functional as per the requirements. Incomplete implementation.	Proper functionality is not achieved, incorrect implementation.	Proper functionality is not achieved. Major bugs/errors found.	Proper functionality is achieved with minor bugs/errors.	The processor is processing instructions correctly.
<u>Reporting the results</u>	Results are not discussed.	All tasks are not reported/discussed.	Few tasks are reported but with some errors/mistakes. Results are briefly discussed.	All tasks are reported but with some errors. Results are discussed but incorrect explanations.	All tasks are appropriately reported and results are thoroughly assessed.

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	